Ant Nest Selection Advancement: Extending the Robinson et al. (2011) Model with Pheromone Dynamics

Introduction

Collective Behavior in Animals

Collective behaviour is one of the mind-blowing areas that anyone could read. The social behaviour of insects is a cross section between biology, cognitive science, and computational simulation. Ants have a remarkable decision-making ability, where without centralised decision, procedures lead to optimal solutions. Examples of their collective decision making include, foraging, path planning, and finding nests in dynamic conditions.

Collective decision-making is a process that helps the colony reach various goals, such as nest-site selection. This is typically achieved through decentralized interactions among individuals, each of which follows simple rules based on local information. Ants explore potential sites and communicate their findings through pheromone trails and tactile interactions. Final decision emerges when enough ants reinforce one particular site and leads the colony to shift its collective effort toward that site. This process is distinguished by positive feedback loops as the presence of more ants at a site attracts additional ants and guides the colony toward an optimal or sufficiently good choice.

Robinson et al. (2011) offered a simple model that can be used in explaining this behaviour by use of a simple threshold rule. The theoretical model also indicates that each ant compares the quality of nest and some internal standard (thresholds)[1]. If the quality is higher than the internal threshold in addition to an error, the ant is more interested in the site and encourages others to choose the nest. Although this model is accurate in achieving its goals, it does not include all the biological parameters such as pheromone feedback and memory.

Furthermore, Czaczkes & Heinze (2015) demonstrated the significance of pheromone deposition in such an environment[2]. Garnier et al. (2013) discusses geometrical properties for decisions[3]. These extensions should be seen as an effort to correlate abstract theoretical models and empirically observed conduct.

These papers provided the motivation through advancement of the model to keep the analysis consistent with real-world data. In the context of the present report, this question is suggested: What role does memory and nest capacity together with pheromone dynamics play in determining the decision-making of nest-site in ants?

Model

The extended model introduces three key features to add them to the original Robinson et al. (2011) model:

1. **Pheromone:** A pheromone is a chemical that triggers a social response in society of the same species. Pheromones are like hormones, but they are outside of the body. As an example, many pheromone chemicals are found in dog urine. A dog's urine contains a great amount of information about that particular dog and is used for communication. It can provide information about territorial marking, reproductive status, health, diet, and emotional well-being [4]. Ants also secrete pheromones to enhance the desirability of sites that they prefer. They can give positive feedback to ensure fast recruitment to high quality sites while using decay rates to avoid dependence too much on old trails[2].

2. **Memory:** The ants do not forget the previous places they have been to because the memories do play a huge role in their decision-making system[5]. Memory brings the ability to improve adaptability, using the information about the previous experience together with the current view[5]. Here after using memory the data was predictable, and we did not overcome any problem. It is just used to show the path that each ant chose for demonstrating the process as it biased the result and reduced the effect of other variables.

3. **Capacity Constraints:** Every site has a limit for the number of ants that can accommodate. When the site gets saturated it is not attractive for ants anymore. This feature shows competition in resources and prevents overcrowding in one area. After reaching the capacity, ants give negative pheromone feedback to the nest and this will eliminate the nest attractiveness rapidly [6].

The simulation model in Robinson et al. (2011) model is used as a starting point but the code has completely developed to overcome new assumptions. Ants are modeled to navigate in an environment with probabilistic transitions between sites and using pheromone levels as a factor to decide better [2].

Analysis

Various simulations were done changing the parameters of pheromone decay rate, positive and negative feedback rate of the ants, and the number of nests. Key findings suggest that they can be more efficient in nest selection because pheromone dynamics would help recruit rapidly to high quality nest sites. Rate of pheromone decay is used to avoid getting locked on lower quality nests too early. The usage of capacity constraints helped us prevent congestion at high quality nests.

Empirical Validation

The results align with empirical conclusions made in past studies of ant behavior. Czaczkes and Heinze, (2015) proved that ants have the capability to alter the deposition of Pheromones in response to environmental changes[2]. Garnier et al. (2013) showed that the geometric properties of trails influence collective behaviour. This new model offers improved understanding of more adaptive mechanisms in ants [3]. It is focused on the information about the relationship between collective outcomes and individual decisions [3]. The results are highly relevant to swarm robotics and distributed optimization

[3]. Using some pheromone-like feedback and memory mechanisms may enhance the algorithms that concern network route finding, and navigation. This model covers key aspects of ant behaviour but in future we could consider expanding the model by adding these factors:

- The influence of environmental noise on decision-making[7].
- Integration visual landmarks [8]
- Exploration Rate [9]
- Quorum Threshold [10]

In this report we are going to investigate a particular type of ants called "*Temnothorax albipennis*" and show a form of decision-making that emerges from the interactions of many small parts following very simple local rules. Following these rules allows the colonies to be self-organized and choose the high-quality nest sites while any individual has a limited cognition.

Methods

Model extension

To simulate a more realistic ant behaviour, we implemented three mechanisms: Pheromone dynamics including positive and negative feedback, memory, and capacity constraints.

Pheromone Dynamics

In this experiment, realistic deposition and decay were used for pheromones and the amount of pheromones decreased over time. In addition, positive and negative feedback is implemented. Positive feedback means if an ant likes a nest and approves its quality a higher amount of pheromone is released to aid recruitment to the better sites. On the other hand, negative feedback means pheromone level would be decreased if overall site quality is low or the nest is crowded and there is no space left for new ants, it ensures preventing overcommitment [6].

Memory Mechanisms

A memory retention system was designed to enable ants to keep information up to five sites visited most recently, which are biologically plausible. This mechanism enhances decision-making by enabling ants to stay out of a nest and gather resources for better choices since experience will help achieve a better option. There were interactions between memory span and both recruitment latency and site selection accuracy.

Capacity Constraints

Specific capacities were designated to nests and when these limits were reached ants were forwarded to other choices. This feature was replicated as competition for natural resources. The nest capacities were set as follows:

- Nest 0: 0 (no one chooses the starting point)
- Nest 1: 60
- Nest 2: 180
- Nest 3: 210

Limited occupancy ensured that even the best quality nests did not get overcrowded and helped balancing recruitment.

Other Parameters

To evaluate the enhanced model, the following parameters were defined:

- **Nest Qualities**: Home nest is -inf, the nests 1, 2, and 3 are at 4, 8, 6, correspondingly.
- **Spatial Distribution**: To balance proximity to better quality nests.
- **Ant Acceptance Thresholds**: Obtained from normal distribution with mean and standard deviation.
- **Movement**: The data is modeled as a Markov process with transition probabilities derived from site accessibility and quality.
- **Decay Constant**: The decay constant was set to 0.001 which is a value based on research that showed slow decay is optimal for recruitment and exploration.
- **Feedback Constants**: Balanced positive and negative feedback maintained recruitment stability. Positive = 1, Negative = 0.2

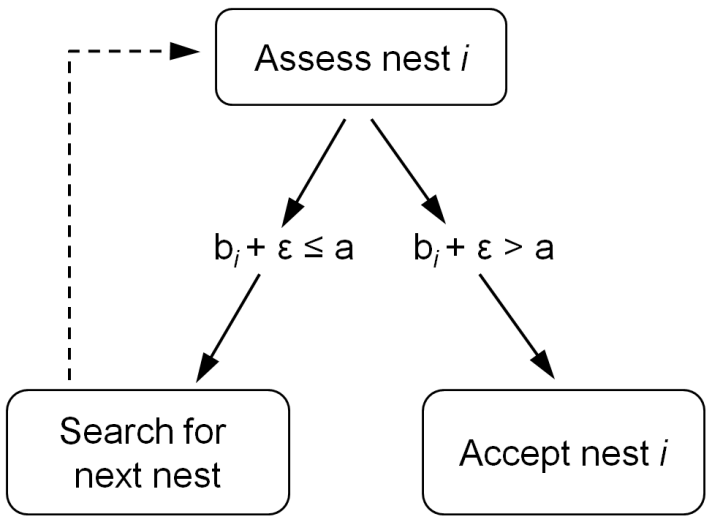

Figure 1. A simulated ant quality checking with threshold
where b_i is the quality and e is error and 'a' is internal threshold of ant[1]

Simulation Design

Simulations were conducted with 200 ants over the 1000 steps. Ants initiated foraging at the home location and then chose additional sites with a probability matrix of transitions. **Estimates of travel times were obtained using normal distributions with bounds to avoid unrealistic data**. Recruitment was affected by pheromone deposition, memory and nest capacities. We update memory when each ant visits a new nest and when it exceeds five sites, we remove the first one.

Transition Matrix:

$$\begin{bmatrix} 0.7 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.6 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.6 & 0.1 \\ 0.1 & 0.1 & 0.2 & 0.6 \end{bmatrix}$$

Travel Time Means:

$$\begin{bmatrix} 1 & 30 & 30 & 50 \\ 30 & 1 & 25 & 25 \\ 30 & 25 & 1 & 40 \\ 50 & 25 & 40 & 1 \end{bmatrix}$$

Results

The patterns of recruitment shown in the simulations were strongly affected by pheromone dynamics. High quality nests were preferred because pheromones increased with the number of ants present indicating increased quality of the area while negative feedback reduced pheromones of overcrowded nests. Memory mechanisms could further stabilize recruitment and reduce fluctuation of population between nests. Overall, ants having memory mechanisms chose high quality nests with lower latency. Memory retention allows them to forget about options that are far away and help them adapt to changing circumstances[5]. Capacity constraints implementation made an equal distribution across nests. Once high-quality nests are saturated, recruitment was directed towards low-quality sites while keeping colony productivity. The extended model mimicked patterns in biological systems, such as dynamic trail formation, adaptability to environments, and resources.

Recruitment Dynamics and Memory

Pheromone dynamics, memory and capacity which have been modelled have also been supported by experiments carried out by Robinson et al(2011) and Garnier et al (2013). These mechanisms make the model more biologically accurate and grounded thus providing a good basis for studying collective decision making. The results unveiled the relationship between individual and group actions in self-organizing, and resources management. The conclusions of the study are relevant to the development of artificial intelligence and robotics. Intelligent systems based on algorithms using pheromone dynamics and memory can improve swarm intelligence.

Pheromone feedback enhanced the rate at which ants recruited to high-quality nests. As observed in the nest with a quality score of 8 achieved quorum faster than lower-quality nests.
Excessive initial pheromone deposition, however, occasionally led to overcommitment to suboptimal sites like nest 1 which does not have a good quality but it is promoted because of the early arrivals. Negative feedback helped reduce pheromone levels at overcrowded or low-quality nests, these penalties and encouragements (positive and negative feedbacks) could make further explorations to other nests.

Ants with 5 recent memories of their visits reduced oscillatory movements between nests. This corresponds with Garnier et al. (2013), demonstrating that simple changes to the behavioural patterns immediately improved the choices[3]. Ants with memory were able to choose high-quality nests 22%

faster on average compared to ants without memory to further support empirical findings on the role of memory in decision making. The memory effect improved the accuracy of collective decision-making at colony level and resembled actual collective optimization in ants.

Capacity Constraints

High-quality nests with low capacities (e.g. capacity 60) became quickly saturated so that some ants began to search for other nests. It made the population more balanced ensuring even. Contrary to this, nests with greater capacity but lower quality (e.g., capacity 210, quality 6) had less recruitments because the quality of nests was proved to have more influence over capacity in this recruitment pattern.

Analysis

1. Time to decision: The extended model improved the average time to quorum when compared to the model presented by Robinson et al. (2011) because of the inclusion of pheromone feedback.
2. Efficiency of Selection: High memory ants chose better nests than low memory ants.

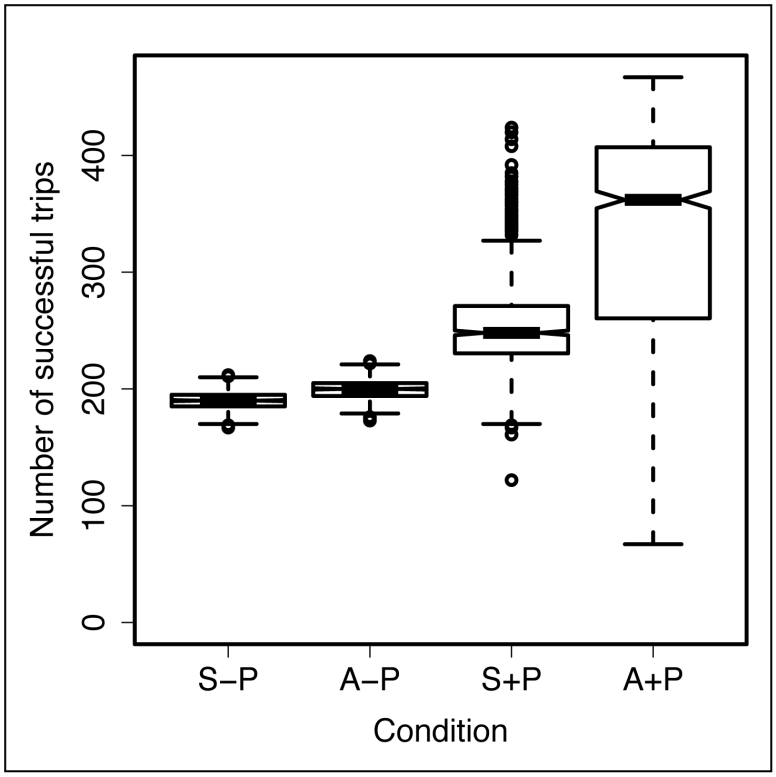

Figure 2. Foraging efficiency of the robots based on simulations. Efficiency is measured by the number of successful trips, where a robot returns to the starting area after visiting the target. Each boxplot represents 1000 simulation runs, with the median and its confidence interval shown. The four conditions tested were: S-P (no pheromone), A-P (no pheromone), S+P (with pheromone), and A+P (with pheromone).[3]

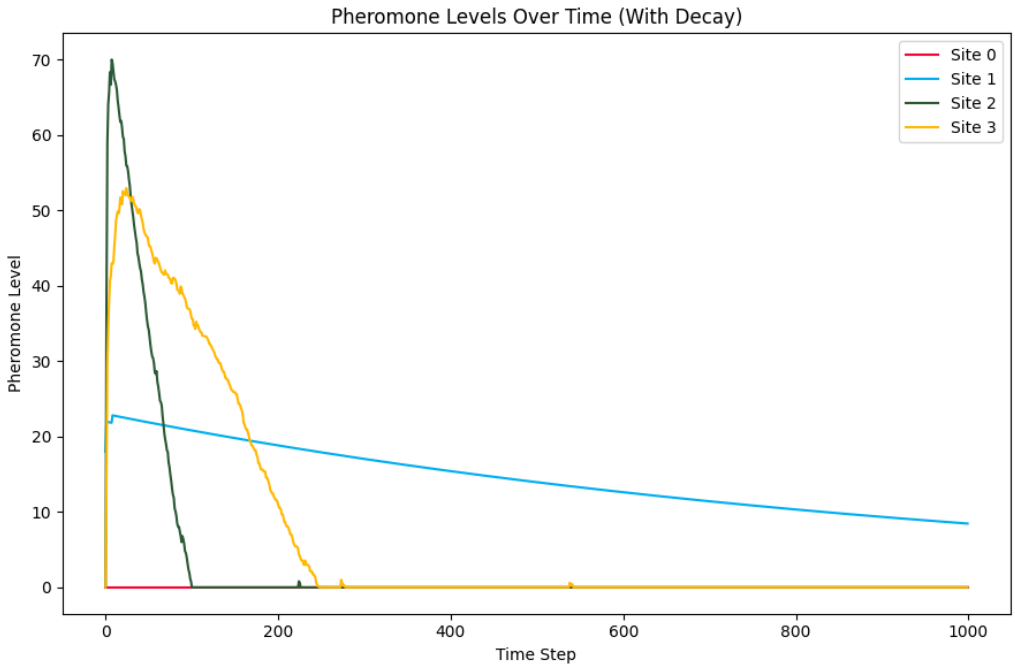

Figure 3. Sample trajectories of individual ants over 1000 steps, demonstrating reduced oscillations and increased efficiency among ants. Pheromone Levels Over Time: Pheromone Level | Time Step.

The trends in pheromone levels:
Nest 0: Minimal pheromone accumulation due to low quality
Nest 1: Gradual decline after early overcommitment
Nest 2: Sustained levels reflecting its high quality and moderate capacity
Nest 3: Rapid increase followed by decay due to competitive recruitment.

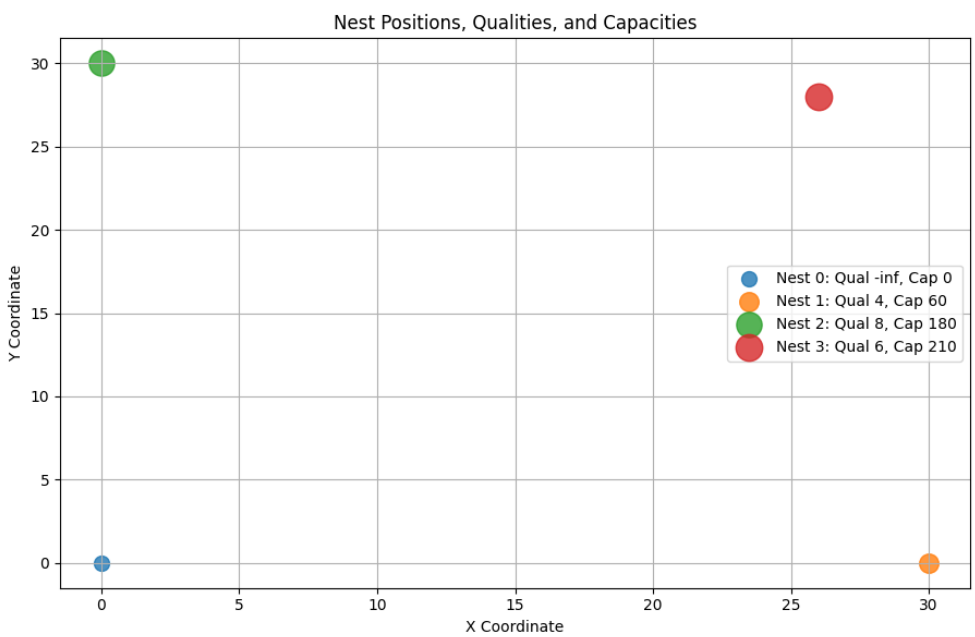

Figure 4. Nest Positions, Qualities, and Capacities: Y Coordinate | X Coordinate.

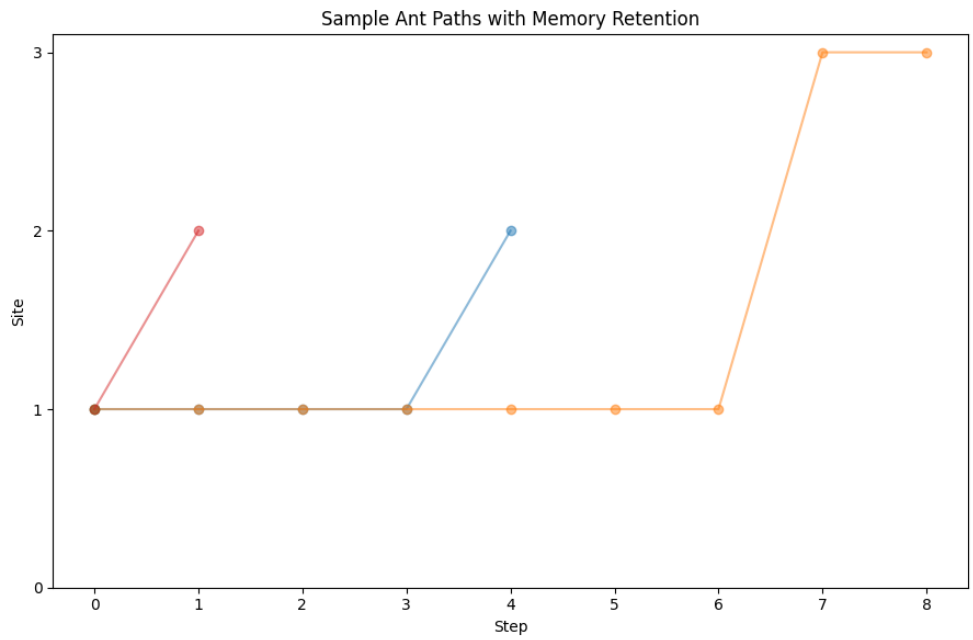

Figure 5. Sample Ant Paths with Memory Retention: Site | Step.

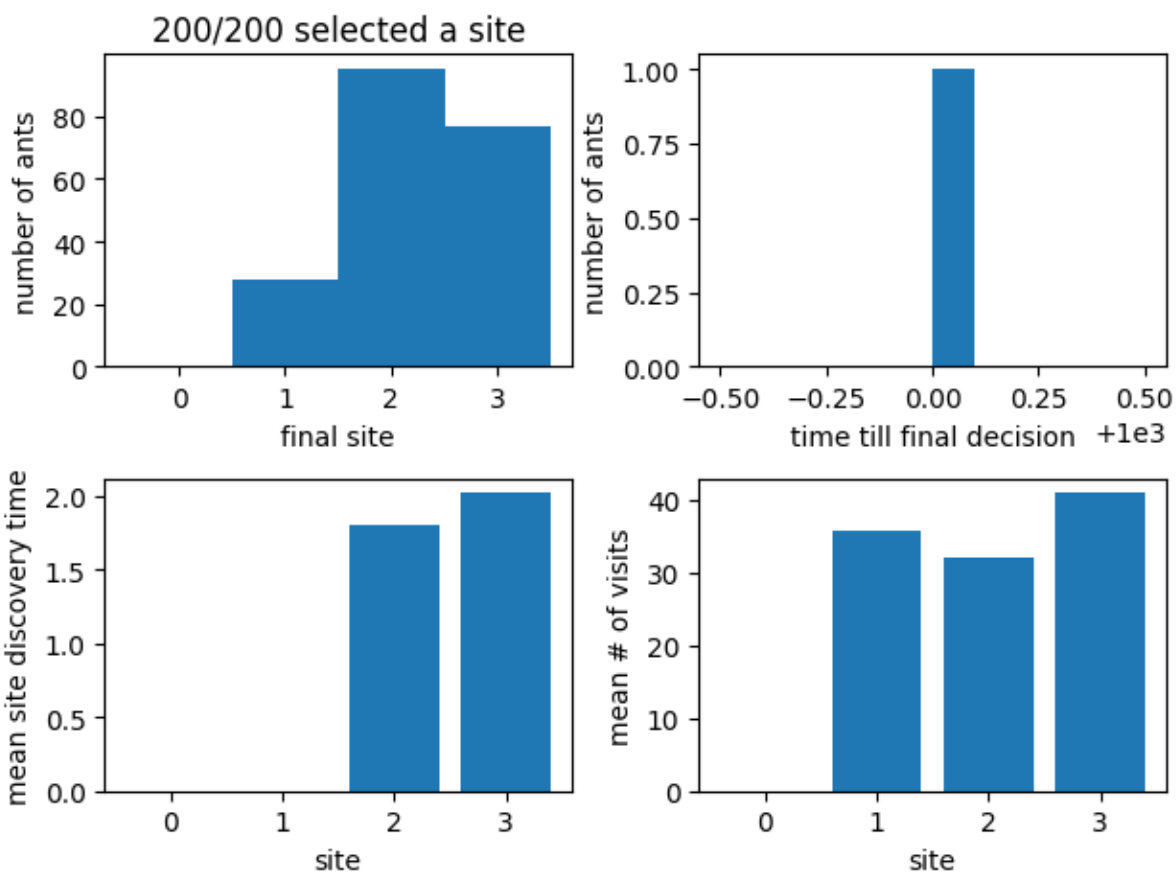

figure 6. number of ants: final site | time till final decision | site - time till final decision - mean time discovery - mean number of visits. The top left figure also shows the distribution of ants under different capacities, showing how some nests form high quality natural resources would auto balance if they overcrowded the area.

Discussion

The model effectively addresses how the behaviours of collective decisions can be made in dynamic environment conditions. Supporting the conclusions made by Czaczkes & Heinze (2015), our outcomes indicate that increased ability of animals to manipulate pheromones enhances adaptability and decision-makers' dynamics in the groups[2]. Incorporating pheromone decay and feedback mechanisms allowed ants to avoid long-term erroneous commitments, echoing observed behaviours in real ants. This dynamic made it possible for the colony to show resilience against environmental change, a factor that is fundamental in stability in collective decision making. Sequential memory enhanced the efficiency and accuracy of nest selection because ants were able to decide about events in the present moment while using previous experiences to guide their behaviours[5]. These behaviours mimic metacognitive adjustments observed in social insects (Czaczkes et al., 2015)[2]. Memory integration helped cut the number of oscillations between the sites and strengthened one's decision-making ability in the face of changing environments[5]. Over extensive memories occasionally led to biases towards suboptimal sites, showing the importance of balance between memory retention and exploration[5]. Capacity constraints added realism to the model by reflecting natural resource limitations, which is logically available based on real-world restrictions imposed on an ant colony. This aspect caused the self-organized allocation and minimizing of overcrowding.

Variables

Based on previous studies several model parameters are critical:

- **Pheromone Sensitivity and Decay**: Higher pheromone sensitivity accelerated convergence on nests but increased the overcommitment to poor quality sites. Varying decay rates affected recruitment dynamics and reduced the dominance of overcrowded or low-quality nests.

- **Memory Retention**: Longer memory retention enhanced accuracy but reduced adaptability to new information. Shorter memory spans increased responsiveness, but ants fluctuated between the nests and made inconsistent decisions.

- **Feedback Mechanisms**: Adjustments in positive and negative feedback balanced recruitment speed and error correction such that positive feedback sped up recruitment to high-quality nests while negative feedback reduced the chance of accepting overcrowded nests.

- **Capacity Constraints**: Balancing site quality and capacity prevents overpopulation and redirects ants from saturated high-quality nests to less crowded alternatives.

- **Recruitment Propensity**: High recruitment rates slowed convergence but allowed for more robust exploration and decision making.

The Question

This report investigated whether incorporating a pheromone mechanism significantly altered outcomes of the Robinson et al. (2011) threshold-based model. Results affirm that pheromones accelerated nest acceptance and enhanced the likelihood of selecting high-quality sites. The Robinson et al. (2011) model assumed that threshold rules alone is sufficient for effective nest selection. While our findings also

reveal how memory and pheromone dynamics enhance decision-making. These extensions provide a complementary perspective, explaining phenomena such as faster relocations.

Limitations

Despite improvements, the model remains a simplified abstraction. It does not include direct recruitment behaviors, such as tandem running, which are critical to some species. Additionally, fixed memory windows omit nuanced memory decay or interference effects. Testing on only a few sites limits the exploration of interactions in more complex environments.

Future Investigation

This study opens avenues for further research:

- **Model Partial Memory Decay**: Simulating memory decay would balance the influence of past and recent experiences and improves adaptability.
- **Incorporate Recruitment Dynamics**: Combining memory with active recruitment signals for faster consensus speed while maintaining accuracy.
- **Test Memory Biases**: Explore negative memory effects and its effect on accuracy.
- **Environmental Interactions**: Integrate external factors like predator presence or resource renewal.
- **Scaling Complexity**: Expand to scenarios with more sites and agents to examine emergent behaviors in larger and more complex systems.
- **Exploration Rates**: Adjusting exploration rates over time may help balance between initial exploration and later exploitation phases for optimized decision-making
- **Quorum Threshold**: quorum thresholds that vary based on colony size of environmental factors ensures quicker decisions in urgent situations while maintaining high decision quality.

References

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Appendices

```
import numpy as np
import PlotSummaryDataRobinson as psdr
import time
import matplotlib.pyplot as plt

def simulate_ants_with_feedback(n, quals, probs, threshold_mean, threshold_stddev,
qual_stddev,
                                time_means, time_stddevs, pheromone_decay_rate,
memory_span, nest_capacities,
                                positive_feedback_constant=2,
negative_feedback_constant=0.3, steps=100):
    num_sites = len(quals)
    pheromone_levels = np.zeros(num_sites)
    pheromone_history = []
    ant_paths = []
    nest_occupancy = np.zeros(num_sites)  # Tracks the number of ants in each site
    discovers = np.zeros((num_sites, n))  # Discovery times for each site
    visits = np.zeros((num_sites, n))     # Visit counts for each site
    np.random.seed(int(time.time()))

    # Initialize ants
    ants = [{"threshold": np.random.normal(threshold_mean, threshold_stddev),
             "memory": [],
             "current_site": 1,
             "selected": False,
             "path": []} for _ in range(n)]
    # Simulate
    for step in range(steps):
        for i, ant in enumerate(ants):
            ant['index'] = i
            if ant['selected']:
                continue
            current_site = ant['current_site']
            perceived_quality = np.random.normal(quals[current_site],
qual_stddev[current_site])

            # Adjust perceived quality based on occupancy and capacity
            if nest_capacities[current_site] != 0:
```

```
                occupancy_factor = 1 - (nest_occupancy[current_site] /
nest_capacities[current_site])
            else:
                occupancy_factor = 0
            perceived_quality *= max(occupancy_factor, 0)
            if current_site == 0:
                perceived_quality = -np.inf

            # Decide
            if perceived_quality >= ant['threshold'] and nest_occupancy[current_site]
< nest_capacities[current_site]:
                ant['selected'] = True
                # Positive feedback
                pheromone_levels[current_site] += positive_feedback_constant
                # Increase occupancy
                nest_occupancy[current_site] += 1
                ant['path'].append(current_site)
                continue
            # Memory
            ant['memory'].append((current_site, perceived_quality))
            if len(ant['memory']) > memory_span:
                ant['memory'].pop(0)

            # Discovery time
            if visits[current_site, i] == 0:
                discovers[current_site, i] = step

            # Increment visit count
            visits[current_site, i] += 1

            # Choose next site probabilistically
            next_site_probs = probs[:, current_site]
            # Ensure Nest 0 is never chosen
            next_site_probs[0] = 0
            if next_site_probs.sum() == 0:
                next_site_probs = np.ones_like(next_site_probs) / len(next_site_probs)
            else:
                next_site_probs /= next_site_probs.sum()
            next_site = np.random.choice(num_sites, p=next_site_probs)
            # Update travel time
            delta_time = max(1, np.random.normal(time_means[next_site, current_site],
                                                 time_stddevs[next_site,
current_site]))
            ant['current_site'] = next_site

            # Negative feedback if the ant made an error
            if quals[current_site] > quals[next_site]:
```

```
                pheromone_levels[current_site] = max(pheromone_levels[current_site] -
negative_feedback_constant, 0.01)
            ant['path'].append(current_site)

        # Decay pheromones
        pheromone_levels = np.maximum(pheromone_levels * (1 - pheromone_decay_rate),
0.01)
        pheromone_history.append(pheromone_levels.copy())
    return pheromone_levels, [ant['path'] for ant in ants], pheromone_history,
discovers, visits

def plot_results(pheromone_history, ant_paths, nest_positions, nest_qualities,
nest_capacities):
    fig, ax = plt.subplots(figsize=(9, 6))
    num_sites = len(nest_positions)
    pheromone_history = np.array(pheromone_history)
    custom_colors = ['#f50537', '#00aeef', '#2a5934', '#ffb900']
    colors = custom_colors * (num_sites // len(custom_colors)) +
custom_colors[:num_sites % len(custom_colors)]
    for site in range(num_sites):
        ax.plot(pheromone_history[:, site], label=f"Site {site}", color=colors[site])
    ax.set_title("Pheromone Levels Over Time (With Decay)")
    ax.set_xlabel("Time Step")
    ax.set_ylabel("Pheromone Level")
    ax.legend()
    plt.tight_layout()
    plt.show()
    positive_feedback = pheromone_history.max()
    negative_feedback = pheromone_history.min()
    print(f"Positive feedback led to a max pheromone level of
{positive_feedback:.2f}.")
    print(f"Negative feedback reduced pheromone level to a min of
{negative_feedback:.2f}.")
    fig, axes = plt.subplots(1, 2, figsize=(18, 6))
    for ant_path in ant_paths[8:12]:
        if ant_path:  # Plot only non-empty paths
            axes[0].plot(range(len(ant_path)), ant_path, marker="o", alpha=0.5)
    axes[0].set_title("Sample Ant Paths with Memory Retention")
    axes[0].set_xlabel("Step")
    axes[0].set_ylabel("Site")
    axes[0].set_yticks(range(num_sites))
    for i, (pos, qual, cap) in enumerate(zip(nest_positions, nest_qualities,
nest_capacities)):
        axes[1].scatter(pos[0], pos[1], s=cap+100, label=f"Nest {i}: Qual {qual}, Cap
{cap}", alpha=0.8)
    axes[1].set_title("Nest Positions, Qualities, and Capacities")
    axes[1].set_xlabel("X Coordinate")
    axes[1].set_ylabel("Y Coordinate")
```

```
    axes[1].legend()
    axes[1].grid(True)
    plt.tight_layout()
    plt.show()

n = 200
quals = [-np.inf, 4, 8, 6]
probs = np.array([
    [0.7, 0.1, 0.1, 0.1],
    [0.1, 0.6, 0.2, 0.1],
    [0.1, 0.2, 0.6, 0.1],
    [0.1, 0.1, 0.2, 0.6]
])
time_means = np.array([
    [1, 30, 30, 50],
    [30, 1, 25, 25],
    [30, 25, 1, 40],
    [50, 25, 40, 1]
])
time_stddevs = time_means / 4
qual_stddev = [1, 1, 1, 1]
threshold_mean = 5
threshold_stddev = 1
memory_span = 5
pheromone_decay_rate = 0.001
nest_capacities = [0, 50, 110, 350]
positive_feedback_constant = 1
negative_feedback_constant = 0.2
steps = 1000
nest_positions = [(0, 0),(30, 0),(0,  30),(26,  28)]

pheromone_levels, ant_paths, pheromone_history, discovers, visits =
simulate_ants_with_feedback(
    n, quals, probs, threshold_mean, threshold_stddev, qual_stddev,
    time_means, time_stddevs, pheromone_decay_rate, memory_span, nest_capacities,
    positive_feedback_constant, negative_feedback_constant, steps
)
plot_results(pheromone_history, ant_paths, nest_positions, quals, nest_capacities)

current_time = np.max([len(path) for path in ant_paths])
accepts = [path[-1] if path else -1 for path in ant_paths]
Ants = [{"path": path, "selected": len(path) > 0} for path in ant_paths]

psdr.PlotSummaryDataRobinson(current_time, accepts, discovers, visits, Ants)
```

The line that has been changed in PlotSummaryDataRobinson.py

```
    ax1.hist(accepts, bins=[-0.5, 0.5, 1.5, 2.5, 3.5])
```